Title: Button design testing

Start date: //_____

Finish date: //_____

Responsible: Yuliia Stepanova

Description

Test goal: increase `begin_checkout` by 5% and not degrade `session with orders`
Hypothesis: We believe that if we make the payment button larger, we will increase the conversion of the beginning of the payment process and not reduce the number of sessions with purchases.

Options

Option A (Control): No changes to the payment button size
Option B (Experimental): With changes, the button size will be increased for the audience.

Data sampling

Target audience: all site visitors.
Traffic split: 50% for option A, 50% for option B

Metrics

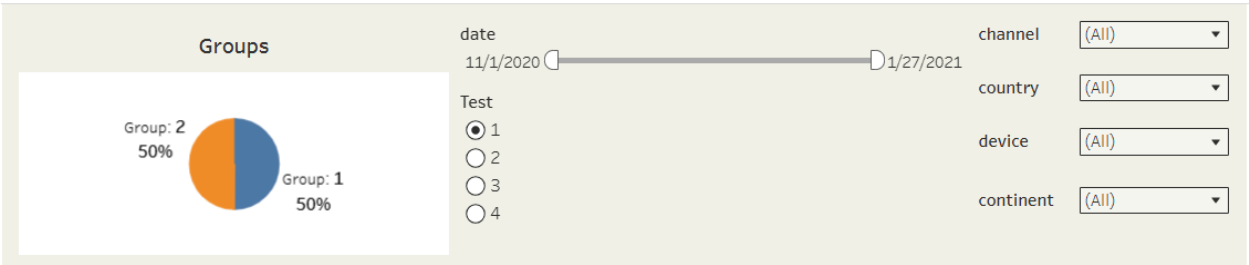
Primary metric: `begin_checkout` / `session`, up 5%
Additional metrics: `session with orders` / `session`, not decreasing

Results

	value		value (%)
	test_group		test_group
event_name	1	2	2
add_payment_info	1,988	2,229	12.12%
add_shipping_info	3,034	3,221	6.16%
add_to_cart	1,395	1,366	-2.08%
begin_checkout	3,784	4,021	6.26%
first_visit	30,596	30,512	-0.27%
new account	3,823	3,681	-3.71%
page_view	191,543	198,050	3.40%
scroll	73,244	73,376	0.18%
select_promotion	1,275	1,323	3.76%
session	45,362	45,193	-0.37%
session with orders	4,514	4,526	0.27%
session_start	45,905	45,649	-0.56%
user_engagement	171,788	179,081	4.25%
view_item	62,335	65,337	4.82%
view_promotion	29,188	29,117	-0.24%

Analyze

Checking the 50% to 50% distribution:



Change in key metrics: In total, the begin_checkout / session metric grew statistically significantly from 8.34% to 8.90%, which is 7%:

	test_group				
event_name	1	2			
begin_checkout	3784	4021			
session	45362	45193			
	8,34%	8,90%	7%		

Evan's Awesome A/B Tools (home):

Sample Size Calculator | **Chi-Squared Test** | Sequential Sampling | 2 Sample T-Test | Survival Times | Count Data

Question: Does the rate of success differ across two groups?

	# successes	# trials	Confidence interval	[clear] [link]
Sample 1:	3784	/ 45362		*8.1% – 8.6%*
Sample 2:	4021	/ 45193		*8.6% – 9.2%*

Verdict:

Sample 2 is more successful

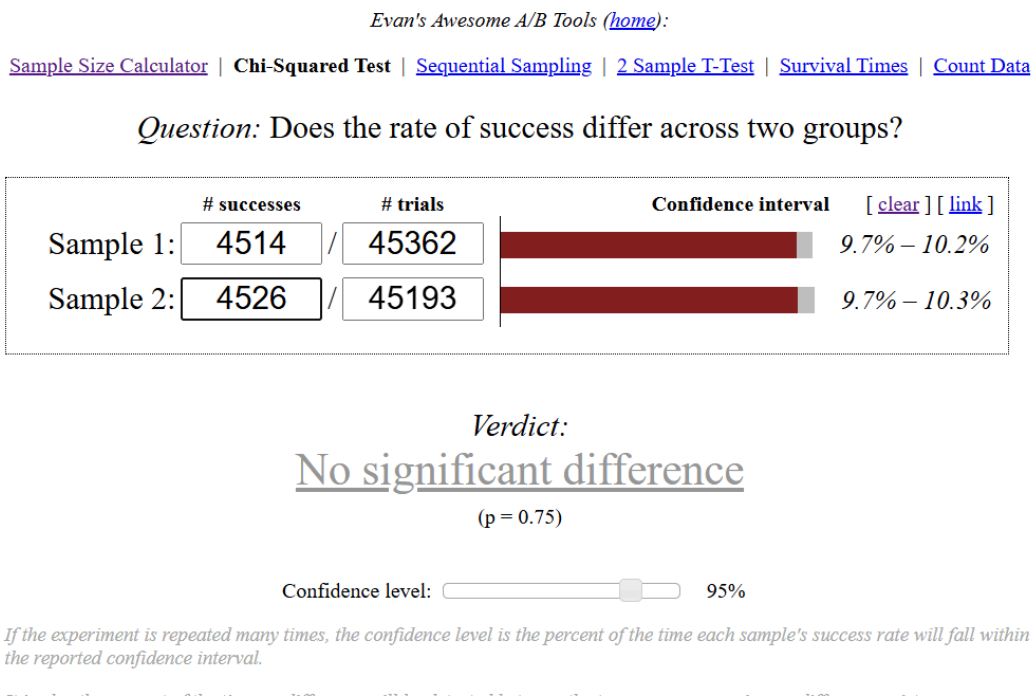
(p = 0.00289)

Confidence level: 95%

If the experiment is repeated many times, the confidence level is the percent of the time each sample's success rate will fall within the reported confidence interval.

Change in additional metrics: The session with orders / session indicator increased by 0.6%, these changes are not statistically significant.

	test_group		
event_name	1	2	
session with orders	4514	4526	
session	45362	45193	
	9,95%	10,01%	0,6%



Changes by segment: In terms of channels, the organic channel gave a negative statistically significant result in the main metric:

	test_group		test_group
event_name	1	2	2
add_payment_info	640	514	-19.69%
add_shipping_info	1,021	922	-9.70%
add_to_cart	494	392	-20.65%
begin_checkout	1,249	1,144	-8.41%
first_visit	12,438	12,305	-1.07%
new account	1,307	1,301	-0.46%
page_view	69,237	69,391	0.22%
scroll	25,561	23,822	-6.80%
select_promotion	442	461	4.30%
session	15,675	15,631	-0.28%
session with orders	1,502	1,534	2.13%
session_start	15,963	15,903	-0.38%

Evan's Awesome A/B Tools (home):

Sample Size Calculator | **Chi-Squared Test** | Sequential Sampling | 2 Sample T-Test | Survival Times | Count Data

Question: Does the rate of success differ across two groups?

	# successes		# trials	Confidence interval	[clear] [link]
Sample 1:	1249	/	15675		*7.6% – 8.4%*
Sample 2:	1144	/	15631		*6.9% – 7.7%*

Verdict:

Sample 1 is more successful

(p = 0.0306)

Confidence level: 95%

If the experiment is repeated many times, the confidence level is the percent of the time each sample's success rate will fall within the reported confidence interval.

In terms of devices, there is a negative trend for *tablets*, there is very little data, but the negative trend is observed for the vast majority of metrics:

	value		value (%)
	test_group		test_group
event_name	1	2	2
add_payment_info	48	31	-35.42%
add_shipping_info	66	49	-25.76%
add_to_cart	37	31	-16.22%
begin_checkout	83	56	-32.53%
first_visit	699	695	-0.57%
new account	90	93	3.33%
page_view	4,829	3,814	-21.02%
scroll	1,751	1,440	-17.76%
select_promotion	29	30	3.45%
session	999	1,009	1.00%
session with orders	98	78	-20.41%
session_start	1,032	1,021	-1.07%
user_engagement	4,010	3,438	-14.26%
view_item	1,692	1,125	-33.51%
view_promotion	660	651	-1.36%

In terms of continents, there is a negative result for Africa:

	value		value (%)
	test_group		test_group
event_name	1	2	2
add_payment_info	19	24	26.32%
add_shipping_info	41	28	-31.71%
add_to_cart	16	14	-12.50%
begin_checkout	54	33	-38.89%
first_visit	325	317	-2.46%
new account	36	38	5.56%
page_view	2,118	1,783	-15.82%
scroll	805	857	6.46%
select_promotion	20	9	-55.00%
session	494	474	-4.05%
session with orders	46	45	-2.17%
session_start	496	486	-2.02%
user_engagement	1,885	1,575	-16.45%

Conclusions and recommendations

Next steps: implement a change in button size for all users except *Organic channel*, *device = tablet* and users from *Africa*. In this case, we will get a larger growth than in total, but the session with orders metric will practically not change.

	value		value (%)
	test_group		test_group
event_name	1	2	2
add_payment_info	1 298	1 680	29,43%
add_shipping_info	1 934	2 247	16,18%
add_to_cart	854	955	11,83%
begin_checkout	2 432	2 820	15,95%
first_visit	17 534	17 610	0,43%
new account	2 427	2 303	-5,11%
page_view	118 010	125 071	5,98%
scroll	45 999	48 129	4,63%
select_promotion	798	834	4,51%
session	28 696	28 613	-0,29%
session with orders	2 920	2 914	-0,21%
session_start	28 932	28 789	-0,49%
user_engagement	106 366	113 716	6,91%

Possible improvements: Look for reasons why the test didn't work for users from Africa, the organic channel, and tablet users.